

A Finite-Gap Reduction for the Haar-Liminf Problem

Candidate partial result for arXiv:2603.17983 and arXiv:2212.11229

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Status: candidate partial result, likely valid. For a symmetric polynomial hypergroup, if the n th Haar weight satisfies $h(n) < 2$, then the Hermitian dual has at least n gaps inside the convex hull of the orthogonalization measure. Hence every polynomial hypergroup whose dual has only finitely many such gaps satisfies $h(n) \geq 2$ eventually, and therefore $\liminf h(n) \geq 2$. The unrestricted infinite-gap case is not settled.

1 The residual open problem

Let $(P_n)_{n \geq 0}$ be a symmetric random-walk polynomial sequence, normalized by $P_n(1) = 1$, and suppose that products have nonnegative linearization

$$P_m P_n = \sum_{k=|m-n|}^{m+n} g(m, n; k) P_k, \quad g(m, n; k) \geq 0, \quad \sum_k g(m, n; k) = 1. \quad (1)$$

Let μ be its orthogonalization probability measure and let

$$h(n) = \frac{1}{\int P_n(x)^2 d\mu(x)} = \frac{1}{g(n, n; 0)} \quad (2)$$

be the Haar weight. Kahler and Szwarc asked whether every such polynomial hypergroup satisfies

$$\liminf_{n \rightarrow \infty} h(n) \geq 2.$$

Their exact list of open problems is reproduced below.

The later paper arXiv:2603.17983 constructs examples with $h(2) < 2$ and thereby fully answers the older pointwise Questions (i) and (iii). It says that its construction gives only partial answers to Questions (iv) and (v), whose asymptotic clauses contain the same liminf issue.

2 The gap obstruction

Write

$$\mathcal{D} = \widehat{\mathbb{N}_0} = \{x \in \mathbb{R} : |P_k(x)| \leq 1 \text{ for every } k \geq 0\}$$

for the Hermitian dual, and let $I = \text{conv}(\text{supp } \mu)$. The standard polynomial-hypergroup facts used below are

$$\text{supp } \mu \subseteq \mathcal{D}, \quad |P_k(x)| \leq 1 \quad (x \in \mathcal{D}), \quad (3)$$

and that P_n has exactly n simple zeros in the interior of I . Call the connected components of $I \setminus \mathcal{D}$ the *dual gaps*.

$$= \{\pm 1\} \cup \{\pm x_n : n \in \mathbb{N}\}.$$

In particular, $\mathcal{X}^b(\mathbb{N}_0)$ is a subset of $\mathbb{R}$, which implies the assertion.

5. OPEN PROBLEMS

We finish our paper with a collection of some open problems:

- (i) Is $h(2) \geq 2$ always true?
- (ii) Is $\liminf_{n \rightarrow \infty} h(n) \geq 2$ always true?
- (iii) Is $h(n) \geq 2$ ($n \in \mathbb{N} \setminus \{1\}$) always true?
- (iv) Is $h(1) \geq 2$ sufficient for $h(2) \geq 2$, $\liminf_{n \rightarrow \infty} h(n) \geq 2$ or $h(n) \geq 2$ ($n \in \mathbb{N}$)?
- (v) Is $0 \in \widehat{\mathbb{N}}_0$ sufficient for $h(2) \geq 2$, $\liminf_{n \rightarrow \infty} h(n) \geq 2$ or $h(n) \geq 2$ ($n \in \mathbb{N}$)?

The questions (i), (ii) and (iii) are motivated by our observations made in Theorem 3.2 (ii) and Theorem 3.3 (v). Concerning (iv) and (v), recall that $0 \in \widehat{\mathbb{N}}_0$ implies at least $h(1) \geq 2$, see the beginning of Section 3.

Figure 1: The residual Haar-liminf question in Kahler–Szwarc, arXiv:2212.11229v3, PDF page 24 (the page is numbered 24 in the text).

- (ii) If $(P_n(x))_{n \in \mathbb{N}_0}$ is as in Theorem 2.1, then $h(2n) < h(2n-1)$ for all $n \in \mathbb{N}$ or $h(2n+1) < h(2n)$ for all $n \in \mathbb{N}$.

Theorem 2.4. *For every $\epsilon \in (0, 1)$, there exist RWPS $(P_n(x))_{n \in \mathbb{N}_0}$ such that nonnegative linearization of products is satisfied and the Haar function h fulfills $h(2) = 1 + \epsilon$.*

Theorem 2.4 gives full answers to the open problems (i) and (iii) presented in [9, Section 4], and it can be seen from the proof below that one also obtains partial answers to the open problems (iv) and (v) of [9, Section 4]. In particular, we shall find examples such that $h(1) > 2$, $h(2) < 2$ and $0 \in \widehat{\mathbb{N}}_0$.

3. PROOF OF THE MAIN RESULTS

The proof of Theorem 2.1 relies on two essential ingredients. The first ingredient is the

Figure 2: Theorem 2.4 and the authors' scope statement in arXiv:2603.17983, PDF page 6.

Theorem 1 (one Haar dip requires many dual gaps). *For every $n \geq 1$ and every $x \in \mathcal{D}$,*

$$P_n(x)^2 \geq \frac{2}{h(n)} - 1. \quad (4)$$

Consequently, if $h(n) < 2$, then $I \setminus \mathcal{D}$ has at least n connected components.

Proof. Apply (1) with $m = n$. The constant coefficient is $g(n, n; 0) = 1/h(n)$ by (2); the other nonnegative coefficients sum to $1 - 1/h(n)$. For $x \in \mathcal{D}$, (3) therefore gives

$$P_n(x)^2 = \frac{1}{h(n)} + \sum_{k=1}^{2n} g(n, n; k) P_k(x) \geq \frac{1}{h(n)} - \sum_{k=1}^{2n} g(n, n; k) = \frac{2}{h(n)} - 1.$$

Suppose $h(n) < 2$. The right-hand side is positive, so no zero of P_n belongs to $\mathcal{D}$. All n zeros therefore lie in $I \setminus \mathcal{D}$. It remains to prove that two zeros cannot occupy the same dual gap.

Assume to the contrary that $r < s$ are zeros of P_n in one connected component of $I \setminus \mathcal{D}$. Since that component is an interval and $\text{supp } \mu \subseteq \mathcal{D}$, the support has no point between r and s . Put

$$q(x) = \frac{P_n(x)}{(x-r)(x-s)}.$$

Then $\deg q = n - 2$, and orthogonality gives

$$0 = \int P_n(x)q(x) d\mu(x) = \int (x - r)(x - s)q(x)^2 d\mu(x). \quad (5)$$

The integrand is nonnegative on $\text{supp } \mu$ and is strictly positive except possibly at the finitely many zeros of q . Because μ has infinite support, the last integral in (5) is strictly positive, a contradiction. Thus the n zeros lie in n different gaps. $\square$

Corollary 2 (finite-gap case). *If $I \setminus \mathcal{D}$ has $G < \infty$ connected components, then*

$$h(n) \geq 2 \quad (n > G).$$

In particular, $\liminf_{n \rightarrow \infty} h(n) \geq 2$.

Proof. The contrapositive of Theorem 1 says that $n > G$ excludes $h(n) < 2$. $\square$

Corollary 3 (two refinements). *The following assertions hold.*

1. *If $0 \in \mathcal{D}$, then $h(n) \geq 2$ for every odd n .*
2. *If $h(n) \leq 2 - \varepsilon$ with $0 < \varepsilon < 1$, then*

$$|P_n(x)| \geq \sqrt{\frac{\varepsilon}{2 - \varepsilon}} \quad (x \in \mathcal{D}).$$

Proof. Symmetry makes P_n odd for odd n , so $P_n(0) = 0$. If $0 \in \mathcal{D}$, this contradicts the strict positivity in (4) whenever $h(n) < 2$. For the second assertion, substitute $h(n) \leq 2 - \varepsilon$ into (4) and take square roots. $\square$

3 Why this does not yet settle the general case

Theorem 1 converts an analytic Haar dip into exact zero-gap complexity. It covers interval duals, finite unions of intervals, and more generally every dual with finitely many gaps in I . It does not bound the number of gaps for an arbitrary polynomial hypergroup. Indeed, the source paper already exhibits polynomial hypergroups with discrete dual $\{\pm 1\} \cup \{\pm x_j : j \geq 1\}$; those particular examples have rapidly growing Haar weights, but they show that infinitely disconnected duals are legitimate.

Eight focused upgrade routes were pursued: a novelty/scope audit; asymptotic analysis of the new explicit family; finite product-positivity optimization; eventually two-periodic recurrence searches; low-order recurrence inequalities; the self-convolution estimate; zero-gap geometry; and an infinite-gap attack. The sixth and seventh routes give the theorem above. The remaining routes neither produce an infinite product-positive counterexample with recurring Haar dips nor forbid arbitrarily many gaps. Thus a full affirmative answer would require new control of infinite-gap duals, while a counterexample would have to realize unbounded zero-gap complexity together with all nonnegative product coefficients.

References

- [1] S. Kahler and R. Szwarc, *Dual spaces vs. Haar measures of polynomial hypergroups*, arXiv:2212.11229v3; J. Approx. Theory 305 (2025), 106099.
- [2] S. Kahler and M. Obermaier, *A new class of orthogonal polynomials*, arXiv:2603.17983 (2026).